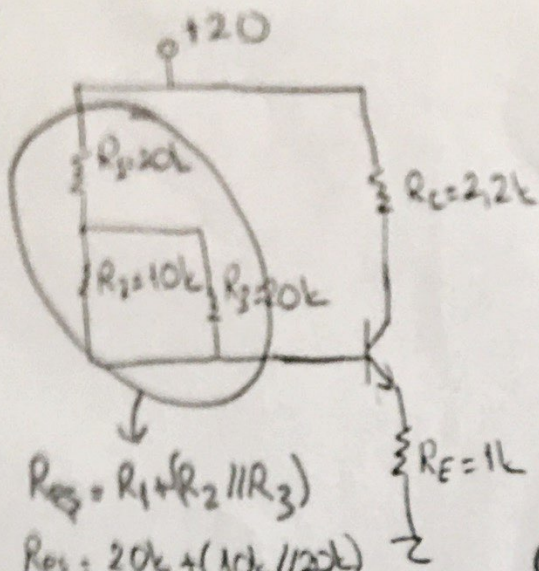


①

a)

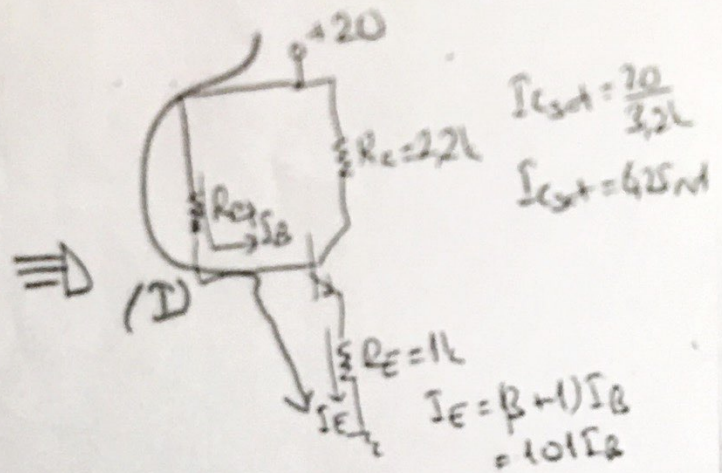


$$R_{eq} = R_1 + R_2 \parallel R_3$$

$$R_{eq} = 20k + (10k \parallel 20k)$$

$$R_{eq} = 26,67k\Omega$$

(2 puan)



$$(I): 20 = I_B \cdot R_{eq} + 0,7 + R_E \cdot I_E$$

$$20 - 0,7 = I_B \cdot 26,67k + 1k \cdot 101 I_B$$

$$I_B = \frac{19,3}{127,67k} = 151,18 \mu A$$

(2 puan)

(4 puan)

$$r_e = \frac{26mV}{6,25mA}$$

$$(r_e = 4,16\Omega)$$

$$\beta r_e = 416\Omega$$

$$I_C = 15,11mA$$

$$I_C > I_{C_{sat}} \text{ old. } I_{C_{sat}} \text{ sat. ver}$$

$$\text{ve } I_C = I_{C_{sat}} = I_E = 6,25mA$$

(Not: eğer $I_C = I_E = 15,11mA$ alıp

$$r_e = \frac{26mV}{15,11mA} = 1,72\Omega \text{ bulmuşlarsa}$$

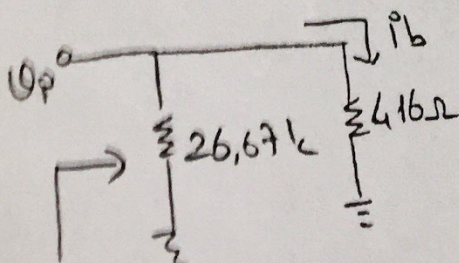
brader Opun

Verelim ana b,c

ve d şıklarından puan kurtarılmış olgusuna.

Yani $\beta r_e = 172\Omega$ ile yaptıkları işlemlere göre değerlendirilm.

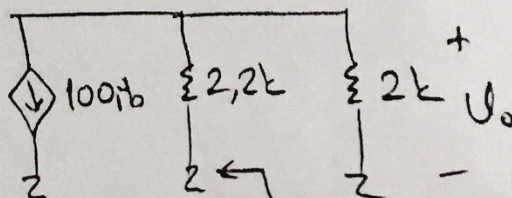
b)



$$R_p = 26,67k \parallel 0,416k$$

$$(R_p = 0,410k\Omega)$$

(3 puan)

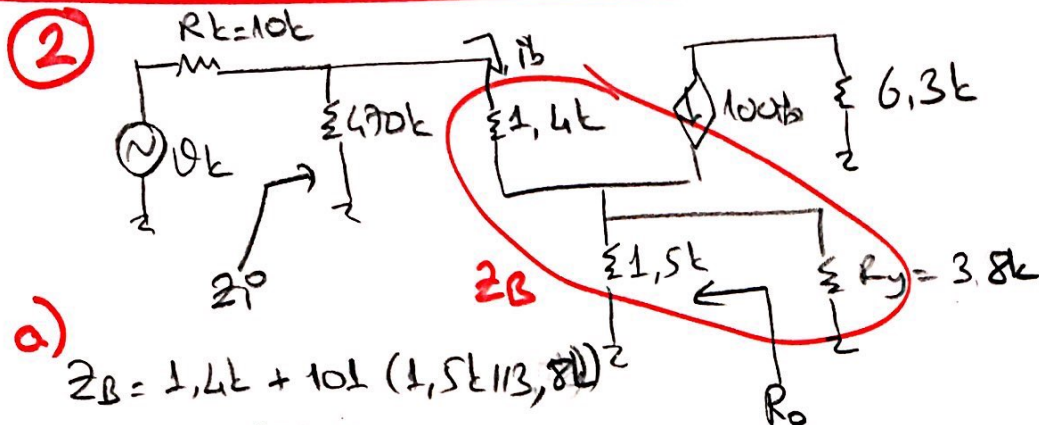


$$R_o = 2,2k\Omega$$

(3 puan)

$$c) A_v = \frac{V_o}{V_i} = \frac{-100 \mu s - (2,2k/12k)}{0,416k \mu s} \Rightarrow \boxed{A_v = -251,83} \quad (6 \text{ pun})$$

$$d) A_{i^o} = -A_v \cdot \frac{Z_i}{R_y} = 251,83 \cdot \frac{0,410k}{2k} \Rightarrow \boxed{A_{i^o} = 51,62} \quad (5 \text{ pun})$$



a)

$$Z_B = 1,4k + 101(1,5k || 3,8k)$$

$$Z_B = 110k \Omega$$

$$Z_P = 470k || 110k$$

$$\boxed{Z_P = 89,13k} \quad (3 \text{ pun})$$

$$R_o = 1,5k || \frac{(10k || 470k) + 1,4k}{101}$$

$$R_o = 1,5k || 0,1108k$$

$$\boxed{R_o = 103,18\Omega} \quad (4 \text{ pun})$$

b)

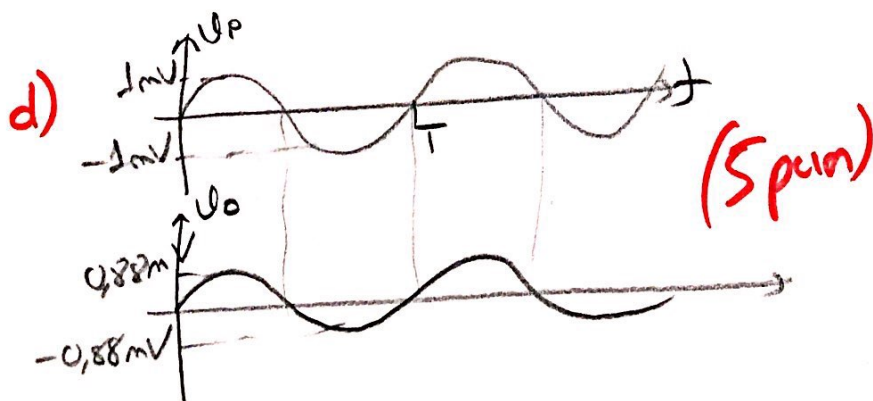
$$A_v = \frac{101 \mu b (1,5k || 3,8k)}{\mu b \cdot Z_B} = \frac{101 \mu b (1,5k || 3,8k)}{101 \cdot 110k}$$

$$\boxed{A_v = 0,986} \quad (6 \text{ pun})$$

$$A_p = -A_v \cdot \frac{Z_P}{R_y} = -0,986 \cdot \frac{86,19k}{3,8k} \Rightarrow \boxed{A_p = -22,75} \quad (4 \text{ pun})$$

c)

$$A_{v_s} = A_v \cdot \frac{Z_i}{R_k + Z_i} = -0,986 \cdot \frac{86,19k}{96,19k} \Rightarrow \boxed{A_{v_s} = 0,88} \quad (3 \text{ pun})$$

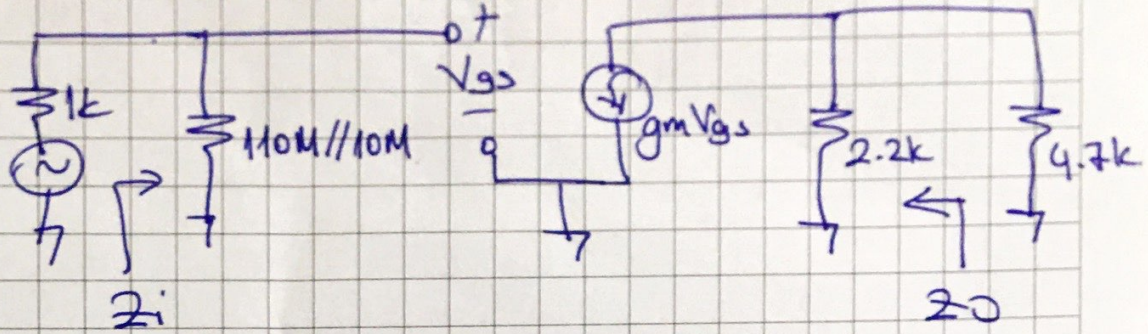


Sgt-2

GÖZÜMLER (3. ve 4. SORU)

3

a)



b) $Z_i = 110M // 10M = 9.17M$ 5 puan

c) $Z_o = 2.2k$ 5 puan

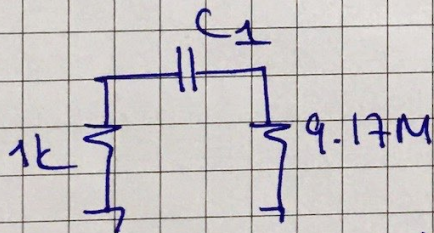
d) $A_v = -g_m (R_D // R_L) = -3.56ms (2.2k // 4.7k)$

$A_v = -5.34$ 5 puan

e) $A_i = -A_v \cdot \frac{Z_i}{R_y} = -(5.34) \left(\frac{9.17M}{4.7k} \right) = 1041.10$ 5 puan

4

a)



$f_1 = \frac{1}{2\pi C_1 \cdot R_{eq}}$

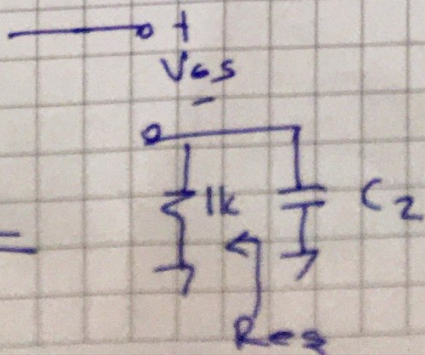
$10Hz = \frac{1}{2\pi C_1 \cdot (9.17M + 1k)}$

$10Hz = \frac{1}{2\pi C_1 (9.28M)} \Rightarrow C_1 = \frac{1}{10 \cdot 2\pi \cdot (9.28)10^6}$

4 puan $C_1 = 0.0017 \cdot 10^{-6} F$

~~3~~

C2 gain



$R_{eq} = R_S // \frac{1}{g_m}$

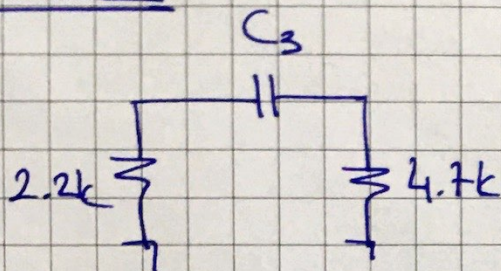
$R_{eq} = 1k // \frac{1}{3.56ms}$

$R_{eq} = 218.78 \Omega$

$$f_2 = \frac{1}{2\pi C_2 R_{ef}} \Rightarrow 10\text{Hz} = \frac{1}{2\pi C_2 \cdot (218.78\Omega)}$$

$$C_2 = \frac{1}{2\pi \cdot 10 \cdot (218.78\Omega)} \Rightarrow \boxed{C_2 = 72.78 \mu\text{F}} \quad (4 \text{ puan})$$

C_3 için

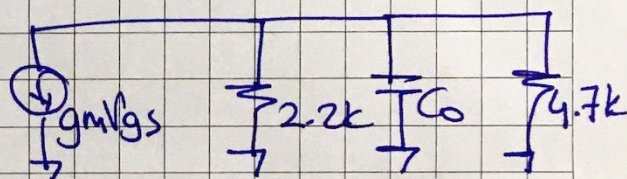
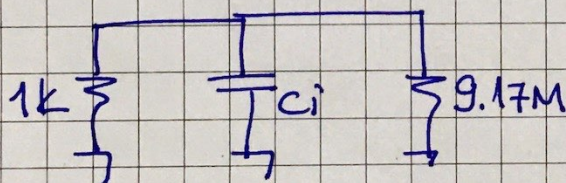


$$f_3 = \frac{1}{2\pi C_3 \cdot R_{eq}}$$

$$10\text{Hz} = \frac{1}{2\pi C_3 (2.2k + 4.7k)}$$

$$C_3 = \frac{1}{2\pi \cdot 10 \cdot (6.9k)} \Rightarrow \boxed{C_3 = 2.3 \mu\text{F}} \quad (4 \text{ puan})$$

b)



$$C_i = C_{gs} + C_{w1} + C_1$$

$$C_1 = C_{gd} (1 - A_v) = 6p(1 - (-5.34))$$

$$C_1 = 38.04p$$

$$C_i = 3p + 10p + 38.04p$$

$$C_i = 51.04p$$

$$f_i = \frac{1}{2\pi \cdot (51.04) \cdot 10^{-12} \cdot (1k \parallel 9.17M)}$$

$$\boxed{f_i = 3.120\text{MHz}} \quad (5 \text{ puan})$$

$$C_o = C_{ds} + C_{w2} + C_2$$

$$C_2 = C_{gd} (1 - 1/A_v) = 6p(1 - 1/(-5.34)) = 7.12p$$

$$C_o = 4p + 3p + 7.12p = 14.12pF$$

$$f_o = \frac{1}{2\pi C_o (2.2k \parallel 4.7k)}$$

$$\boxed{f_o = 7.52\text{MHz}} \quad (5 \text{ puan})$$

$$\boxed{f_{yst} = 3.12\text{MHz}} \quad (3 \text{ puan})$$